Feature Joint Learning for SAR Target Recognition

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***Abstract*— The features employed for synthetic aperture radar (SAR) target recognition have evolved from traditional SAR target geometric features and pattern features to modern deep features, indicating a trend of increasing recognition accuracy but decreasing feature interpretability. Therefore, the fusion of multidimensional features has been investigated by many researchers. Existing feature fusion methods typically involve simple concatenation or addition of geometric features and pattern features with deep features, or directly incorporating them into deep networks. However, such fusion methods mentioned above inadequately consider the potential conflicts between features and hard to fully exploit multidimensional features. To solve the above problem, a multidimensional feature joint learning framework (MFJL-Framework) that serves the SAR target recognition task is proposed in this article, which consists of three models. Specifically, the SGC-GA-Model can select pattern features for SAR targets based on geometric feature constraints, the Global and local Feature Information interaction Capture model (GFIC-Model) can select deep features with high-level abstract semantics, and the MFFS-Model can complement and fuse these two types of features to maximize the utilization of feature information. Experiments and comprehensive ablation studies on four datasets, namely OpenSARShip-1.0, FUSAR-Ship, MSTAR-T72Variants, and SAR-AIRcraft-1.0, collectively demonstrate that the recognition performance of our proposed FJL-Framework outperforms the current state-of-the-art methods.**

***Index Terms*— Feature joint learning, pattern features, SAR target recognition, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is a kind of active aviation and aerospace ground detection system that achieves coherent imaging of targets by transmitting electromagnetic pulse and receiving backscattered information from the ground surface, which has many advantages, such as all-weather, all-day, multiband, multipolarization, and strong penetration capability. In general missions, SAR can be used for environmental monitoring, urban planning, mapping [1], [2], [3], etc.; in special missions, SAR can achieve the detection and identification of key targets, such as aircraft, ships, tanks, and vehicles [4], [5], [6], [7].

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Effective feature extraction and utilization in SAR target recognition tasks is a crucial factor in determining the SAR target recognition capability. In a long period of research, feature utilization in SAR target recognition can be broadly classified into two types: 1) based on traditional geometric and pattern features and 2) based on advanced semantic features extracted by modern deep models.

The most notable feature of geometric features and pattern features is the extraction of target features by manual methods, which usually use rigorous mathematical theoretical models or draw on the experience of well-established experts to analyze the features of different targets in SAR images. Geometric features are usually obtained directly from the pixels of the SAR target image, while pattern features are often obtained by matrix transformation or domain transformation.

Before the advent of deep models, a variety of geometric features and pattern features have been designed to achieve the task of SAR target recognition in the field of remote sensing [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24]. For example, Song et al. [11] designed a gradient histogram oriented SAR target recognition feature (SAR-HOG) and used a supervised discriminative dictionary learning approach to simplify and optimize the recognition task. Akbarizadeh [12] proposed an algorithm to extract texture features using wavelet transform. Yang et al. [13] fused the three types of ships' structure, strength, and density features to complete the recognition task.

However, with the continuous development of SAR imaging technology, the feature information related to SAR targets grows massively, and the resolution of SAR images is also increasing. As the migration generalization ability of traditional geometric features and pattern features is not strong enough to cope with the many challenges in SAR target recognition in complex scenes. Moreover, their classification accuracy is often lower than that of the high-performance deep models.

In recent years, deep learning has been widely used in the field of SAR target recognition. Compared with the traditional way of using geometric features and pattern features for recognition, deep learning algorithms are able to automatically extract high-level abstract semantic features from images layer by layer for SAR target recognition tasks, and their end-to-end training paradigm simplifies the work design process [14], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36]. For example, to address the problem of labeling SAR targets with too small amount of data, Lu and Li [14]

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proposed a convolutional neural network (CNN) combined with migration learning applied to SAR target recognition method for small datasets. He et al. [25] proposed a ternary CNN framework using dense convolutional networks to extract better deep feature extraction.

However, the deep models applied to SAR target recognition tasks are often directly migrated from the optical domain, without being adapted to the specific needs of the SAR target recognition domain. Second, deep features with high-level abstract semantics are also associated with "black-box effect" and non-interpretability, and cannot be applied in many special scenarios.

By comparing the advantages and disadvantages of traditional geometric, pattern features, and deep features in the SAR domain, we realize that feature fusion of geometric, pattern features, and deep features is a highly valuable strategy for SAR target recognition tasks. This fusion effectively utilizes the strengths of both, while mitigating their respective shortcomings. Pattern features typically focus on the specific physical and geometric attributes of SAR targets, offering strong interpretability and robustness, whereas deep features are capable of capturing more abstract and complex image patterns, establishing connections between the target and its overall background. Combining these can provide more comprehensive information, improve the accuracy of target recognition, and help the model maintain good performance under a variety of complex conditions and environments.

The above ways of utilizing different features have also triggered more and more scholars to think, so multidimensional feature fusion has started to be widely researched in recent years [37], [38], [39], [40], [41], [42], [43], [44]. The existing researches can be broadly classified into two ways: 1) fusion research on extracted multiscale features in deep models. For example, Bai et al. [37] proposed a novel lightweight multiscale deep network with a spatial pyramid attention model to extract fine-grained feature information. Qi et al. [38] proposed a compression-excited visual transformer model based on local feature fusion for local feature fusion and channel modeling, balancing the contribution of each feature. 2) Introducing traditional geometric features and pattern features into the deep model for feature fusion research. For example, Zhang et al. [39] proposed a learning network that combines histogram of oriented gradients (HOG) and deep features. Zheng et al. [40] proposed a new multifeature cooperative fusion network with depth supervision (MFCFNet), which effectively fuses SAR target's traditional features and deep features.

However, there are some potential risks in the current feature fusion methods. In the former work, the deep features with abstract high-level semantics are still the main research target, ignoring the characteristics and advantages of traditional geometric features and pattern features in SAR field. Besides, due to their weak interpretability, they cannot be trusted in a large number of real-world application scenarios.

Among the latter, traditional pattern features are input into deep models for joint training. However, this will allow the pattern features to be extracted as higher-level abstract features, destroying their original SAR target features. In addition, some approaches employ direct connections with the fully connected (FC) layers of deep models or add pattern features to achieve integration. Although there is a certain improvement in the final target recognition accuracy, simply splicing or adding the features with different physical meanings will produce redundant features, and there is also a mutual exclusion effect between different features. Meanwhile, such an approach will also make the optimization plane more complex, and there is no guarantee that there will be positive gains from the introduction of traditional pattern features.

The feature joint learning method presented in this article is specifically designed for SAR target recognition scenarios and is based on the principle of feature fusion—selecting the best from the best rather than merely combining them indiscriminately. This is the fundamental goal of the feature joint learning framework (FJL-Framework). The framework ensures that the combined multidimensional features consistently provide positive gains for SAR target recognition tasks, enabling the retention of higher recognition capabilities and the potential for increased accuracy with features of lower dimensions.

Therefore, an FJL-Framework for multidimensional features fusion of SAR targets is proposed, which aims to allow traditional pattern features and deep features with high-level abstract semantics to perform complementary self-learning to prefer single-branching features first, and then jointly learn them to finally complete the recognition task. Three models are proposed to ensure the improvement of recognition accuracy. These include: 1) SAR target Subspace-level feature selection model with Geometric feature Constraints based on improved Genetic Algorithm (SGC-GA-Model); 2) Global and local Feature Information interaction Capture model (GFIC-Model); and 3) Multidimensional Feature Joint Learning model (MFJL-Model).

The first two models perform branching learning process for traditional pattern features and deep features with abstract high-level semantics targeted preference respectively, and the third model realizes the joint learning of the two, which is synergistically applied to the SAR target recognition task.

The main contributions of this article are as follows.

1) The multidimensional FJL-Framework for SAR target recognition tasks is introduced for the first time, facilitating the joint learning of traditional pattern features and deep features. This significantly enhances the performance of SAR target recognition and has been effectively validated across multiple different types of datasets. Unlike conventional feature fusion methods, this optimized framework is based on a thorough analysis of existing technologies and targeted improvements. It transcends simple feature concatenation or addition by considering the interdependence and complementarity of features at every step of the model.
2) Within the FJL-Framework, a novel subspace-level feature selection model for pattern features in the SAR domain, dubbed the SGC-GA-Model, is proposed. This model utilizes a genetic algorithm (GA) optimized by geometric feature constraints to select feature subspaces. It aims not only to enhance the classification performance of pattern features but also to incorporate

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geometric constraints of SAR images to preserve features that are more conducive to target recognition and reflect the intrinsic physical characteristics of SAR targets, thereby achieving feature dimensionality reduction.

3) Within the FJL-Framework, for deep features in the SAR domain, the GFIC-Model has been proposed. This model implements dynamic feature fusion across scales and mapping layers within a deep network by introducing interaction layers. It facilitates cross-layer and cross-scale information interactions, enhancing the model's adaptability and generalization ability in complex scenarios.
4) Within the FJL-Framework, addressing the effective fusion of pattern features and deep features, the MFJL-Model has been introduced. This model achieves meticulous integration of feature hierarchies through a structured design. It not only synchronously optimizes traditional pattern features and deep features but also coordinates their interactions through a unified joint learning process.

II. Related Work

In this section, the previous related works will be reviewed, which is divided into three main directions: 1) methods based on traditional geometric features and pattern features; 2) advanced semantic feature methods based on modern deep model extraction; and 3) methods for multidimensional feature fusion.

A. Geometric and Pattern Feature Extraction

Traditional geometric features and pattern feature methods usually use rigorous mathematical theoretical models or with the help of mature expert experience to analyze the features of different targets in SAR images, which have sufficient mathematical arguments and interpretability, and are also the earliest methods applied in the field of SAR target recognition.

In previous studies in 2010 and earlier, Wang et al. [15] used SAR target shape constraints for identification. Dalal and Triggs [16] demonstrated experimentally that a grid of HOG descriptors significantly outperforms the set of detected features. Margarit and Mallorqui [17] showed that SAR target geometries can be described by specific combinations of permanently polarized scatterers (PePS).

Between 2011 and 2020, Zhao et al. [18] proposed a target recognition scheme based on multimodal feature evaluation measures such as geometric structural features, transformation features, or local invariant features. Zhang et al. [19] extracted the smallest outer rectangle of the target object and used its derived features to achieve good performance in the recognition task. Wang et al. [20] designed a geometric and backward scattering feature based on different target types in a hierarchical classifier. In the study by Wu et al. [21], for high-resolution SAR images where detailed information about the target structure can be observed, feature vectors are estimated by computing the average of kernel density estimation, three structural features, and the average backscattering coefficient. Lin et al. [22] improved the SAR-HOG features proposed by Song et al. [11] to achieve dimensionality reduction by using stream learning and is co-trained under the dictionary learning (TDDL) framework for collaborative training.

From 2021 to present, Xu and Lang [23] proposed a Geometric Transfer Metric Learning (GTML) method, which implements Discriminative Information Preservation (DIP), Geometric Structure Preservation (GSP), and Domain Shift Synchronized Processing (DS) to improve the classification performance of SAR ships. Fan et al. [24] proposed a multiscale single-gene spatial feature extraction strategy using a regional Euclidean (L2) paradigm to extract features of single-gene signals.

Traditional geometric features and pattern features have rigorous mathematical arguments, as well as being hand-designed, and often have good results for specific recognition problems. Before the advent of deep learning, it was the first choice of many scholars in the field of SAR target recognition.

B. Deep Feature Extraction

Compared with the traditional approach of using geometric features and pattern features for recognition, deep learning models can automatically extract high-level abstract semantic features from targets layer by layer [45], and more and more scholars are applying them to the field of SAR target recognition.

From 2015 to 2020, for the SAR target recognition task, Li et al. [26] proposed a new dense residual network model with a loss function to maximize intra-feature class tightness and inter-feature class separation. Aiming at the problems of parameter redundancy of FC structures and underutilization of channel information in CNN, Huang et al. [27] proposed group compression-excited sparsely connected convolutional networks (GSESCNNs). De Laurentiis et al. [28] proposed Capsule Neural Networks (CapsNets) to improve the use of spatial information, especially the spatial dependency between features. Aiming at the problem of low learning efficiency of CNN models under small sample conditions, Yan et al. [29] introduced a prototype-based metric learning method, which facilitates the extraction of discriminative features.

From 2021 to present, to address the problem of uneven distribution of SAR targets and the inability of traditional CNN models to identify rare types, Zhang et al. [30] proposed a combination of dynamically weighted sampling and soft threshold shifting. Wang et al. [31] proposed a semi-supervised learning framework containing consistent augmentation rules to alleviate the need for labeled data. Zeng et al. [32] proposed a new CNN approach for further accurate SAR target classification. Zhu et al. [33] proposed a small-size convolutional metamer for SAR target identification using a progressive neural architecture search algorithm to find the optimal base model. Aiming at the problem that CNN cannot make full use of SAR target feature representation, He et al. [41] proposed a multiscale framework to fuse convolutional features of different spatial scales. Wang et al. [42] used

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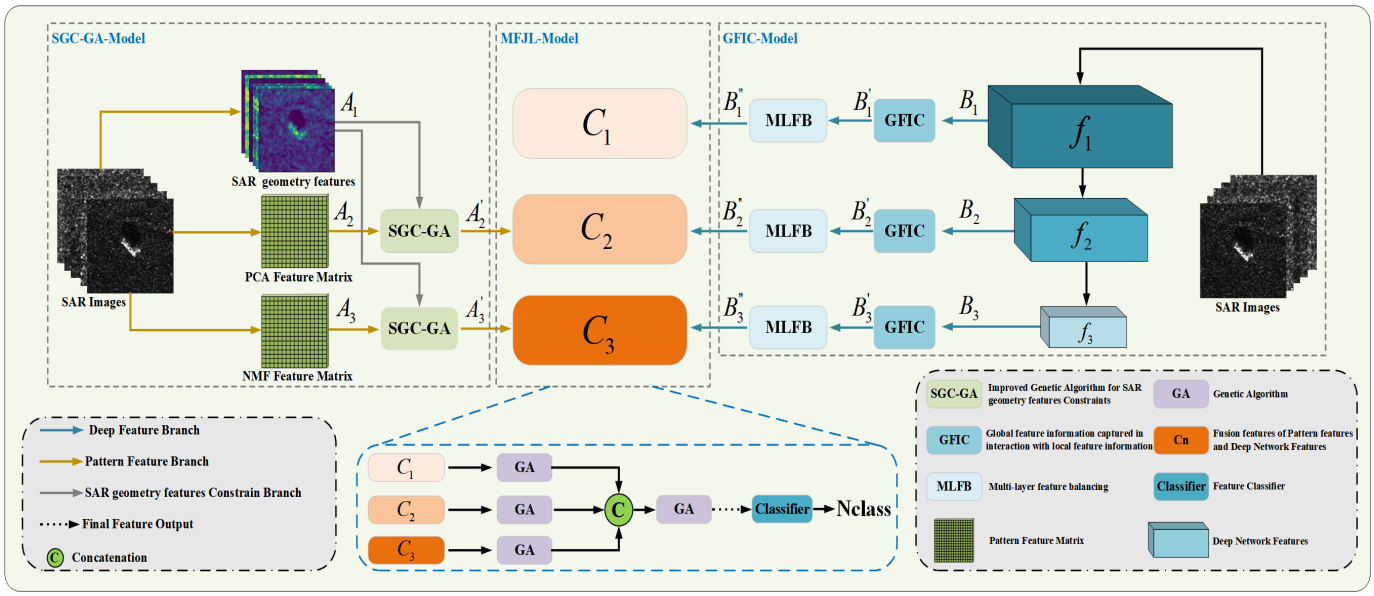


Fig. 1. Overall FJL-Framework. SGC-GA-Model: Subspace-level feature preference model of improved GA based on SAR target geometric features constraints. GFIC-Model: Global feature information and local feature information interaction capture model. MFJL-Model: Multidimensional feature fusion and selection.

multiscale feature focusing and adaptive selection of effective feature scales to achieve the final accurate recognition.

The deep model's end-to-end training paradigm simplifies the work design process and improves productivity, but it also comes with non-interpretability and cannot be directly applied in many scenarios.

C. Multidimensional Feature Fusion

As the scenarios of SAR applications become more and more complex, more and more research has started to use multidimensional features in SAR target recognition tasks.

Feng et al. [43] proposed a recognition method based on the fusion of component models with deep learning algorithms, which fuses local features of component models computed by attribute scattering centers with global deep features. Zhang and Zhang [44] proposed a polarization fusion network (PFGFE-Net) for geometric feature embedding. Zhou et al. [46] proposed the DiffDet4SAR network, which incorporates the Scattering Feature Enhancement (SFE) module, to reduce the clutter intensity of target background and enhance target saliency. This achieves superior performance in challenging aircraft target detection tasks.

Geometric and pattern features lack the capacity to capture more complex patterns and higher-level semantic information, resulting in insufficient discriminative power and intraclass variability among targets. Deep features, automatically learned by deep learning models, can capture advanced abstract semantic information, but they lack interpretability and may face generalization issues without sufficient training data. To address these deficiencies in single-branch features, the SGC-GA-Model and GFIC-Model were proposed.

Furthermore, addressing the issue that feature fusion strategies often rely on simple addition or concatenation of features without fully utilizing their potential, the introduced FJL-Framework offers a more complementary approach to handle different features, considering their complementarity and potential redundancy. It integrates the intuitive representation of geometric features, the detailed capture of pattern features, and the advanced semantics of deep features. Such fusion enables the model to consider multiple aspects of target recognition comprehensively. The potential advantage of this approach lies in its ability not only to enhance recognition accuracy but also to improve the model's adaptability to varying conditions and variations.

III. Proposed Method

A. Overall FJL-Framework

The overall flow of the proposed SAR target recognition MFJL-Framework is shown in Fig. 1. It consists of three parts, namely SGC-GA-Model, GFIC-Model, and MFJL-Model. The three models start from the optimization of the traditional pattern features, a deep feature optimization with high-level abstract semantics, and a multilayer feature joint learning to achieve the overall strategy.

Among them, SGC-GA-Model is a subspace-level feature preference model based on the constraints of SAR target geometric features, which enhances the ability to express the contour and local detail features of the central target object, and at the same time, it can achieve subspace feature reconstruction with good interpretability, and the process will be detailed in Section III-B. In FJL-Framework, A_1, A_2, and A_3 represent the SAR target geometric features, PCA features of SAR image, and NMF features of SAR image, respectively. A_2' and A_3' represent the subspace-level preferred features of PCA and NMF features under the constraint of SAR characteristic A_1, respectively. The SGC-GA represents the improved GA with the constraints of SAR target geometric features.

GFIC-Model is a global feature information and local feature information interaction capture model, which can

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Algorithm 1 FJL-Framework Processing

Input: A set of SAR images X.

begin

  begin

    do: SAR-Related Features extraction.

    $A_1 = \begin{cases} \mathcal{F}_{GLCM}(X) \\ \mathcal{F}_{SNR}(X) \\ \mathcal{F}_{PSNR}(X) \end{cases}$

  end

  begin

    do: SAR image PCA and NMF feature extraction.

    $A_2 = \mathcal{F}_{PCA}(X)$

    $A_3 = \mathcal{F}_{NMF}(X)$

  end

  begin

    do: SGC-GA-Model.

    $A_2' = \mathcal{F}_{SGC-GA}(A_2, A_1)$

    $A_3' = \mathcal{F}_{SGC-GA}(A_3, A_1)$

  end

  begin

    do: GFIC-Model.

    $B_1' = \mathcal{F}_{GFIC}(B_1)$

    $B_2' = \mathcal{F}_{GFIC}(B_2)$

    $B_3' = \mathcal{F}_{GFIC}(B_3)$

    then:

    $B_1'' = \mathcal{F}_{MLFB}(B_1')$

    $B_2'' = \mathcal{F}_{MLFB}(B_2')$

    $B_3'' = \mathcal{F}_{MLFB}(B_3')$

  end

  begin

    do: MFJL-Model.

    $C_1 = \mathcal{F}_{MFJL}(B_1'')$

    $C_2 = \mathcal{F}_{MFJL}(A_2', B_2'')$

    $C_3 = \mathcal{F}_{MFJL}(A_3', B_3'')$

  end

end

Output: $Classification\{GA[GA(C_1) + GA(C_2) + GA(C_3)]\}$

effectively improve the integration ability of global feature information and local feature information for recognizing the target, and the process of its action will be detailed in Section III-B. In the process of deep feature extraction, the neural network model is taken only as a deep feature extractor. In this article, the ResNet-18 [47] is adopted as base model to extract the shallow, medium, and deep layers of features as B_1, B_2, and B_3, respectively, and then capture the interaction between the global feature information and local feature information model to get the feature layers as B_1', B_2', and B_3', respectively. Finally, the corresponding feature layers B_1'', B_2'', and B_3'' are obtained through the MLFB feature balancing process.

The overall algorithmic process of the FJL-Framework is presented as Algorithm 1, with the specific steps outlined as follows.

Step 1: Implementation of SAR target geometric features extraction such as GLCM matrix, SNR, and PSNR. The input is SAR image X, and the output is feature layer A_1; more details will be introduced in Section III-B.

Step 2: Implement SAR image PCA and NMF feature extraction to extract the PCA and NMF features of the SAR image, the input is the SAR image X, and the output is the PCA feature layer A_2 and the NMF feature layer A_3, respectively; more details will be introduced in Section III-B.

Step 3: Implement SGC-GA-Model, which selects PCA features with NMF features under the constraints of SAR target geometric features. The inputs are feature layers A_1, A_2 and A_1, A_3, respectively, and the outputs are feature layers and A_2', A_3'; more details will be presented in Section III-B.

Step 4: Implement GFIC-Model and extract the depth feature layers B_1, B_2, B_3 using the depth feature extractor, and input them into the GFIC module, and the outputs are B_1', B_2', B_3'. The feature layers B_1', B_2', B_3' are then input to MLFB for feature dimension balancing, and the outputs are B_1'', B_2'', B_3'', and more details will be introduced in Section III-C.

Step 5: Implement MFJL-Model to jointly learn the feature layers A_2', A_3' and B_1'', B_2'', B_3'' from Steps 3 and 4, with outputs C_1, C_2, C_3, and more details will be presented in Section III-D.

Step 6: Finally the final classification discriminant is output using the integrated random forest-based classifier.

B. SGC-GA-Model

1) SAR Target Geometric and Pattern Features Extraction: SGC-GA-Model is a subspace-level feature preference model based on SAR target geometric features constraints, a separate learning branch for pattern features. The overall process is shown in Fig. 2, which is proposed to enhance the positive gain of pattern features during SAR image recognition. By introducing the SAR target geometric features constraints, the ability to express the contour and local detail features of the central target object is greatly enhanced, while the feature dimensions required for classification are reduced. The sub-space level of the preferred scheme allows the selected features to achieve image reconstruction, enables visual comparison of key feature parts, and enhances the feature interpretability in the SAR target recognition process.

a) Pattern features extraction: For the SAR target dataset used in the experiments, the PCA features and NMF features are extracted to be adjusted to the same size of the square array for both the training set and the test set before feeding them into the preferred scheme.

For a SAR target dataset $X \in R^{m \times n}$ (m samples with a total of n-dimensional features), where each sample is $x^{(i)} = [x_1^{(i)}, x_2^{(i)}, \ldots, x_n^{(i)}]$, the first 100 dimensions of the PCA feature matrix $V \in R^{100 \times n}$ and the NMF feature matrix $H \in R^{100 \times n}$ are extracted while keeping the base matrices V_{PCA} and H_{NMF} of both.

b) Geometric features extraction: Geometric features extract several features in the GLCM matrix with signal-to-noise ratio (SNR) and peak signal-to-noise ratio (PSNR) features.

Gray-Level Co-occurrence Matrix (GLCM) is used to obtain second-order statistical texture features of images and is a widely used statistical technique developed specifically for processing remote sensing data [48]. The technique is divided into two steps: First, the original image is converted into a grayscale [49] image. Second, the spatial features are

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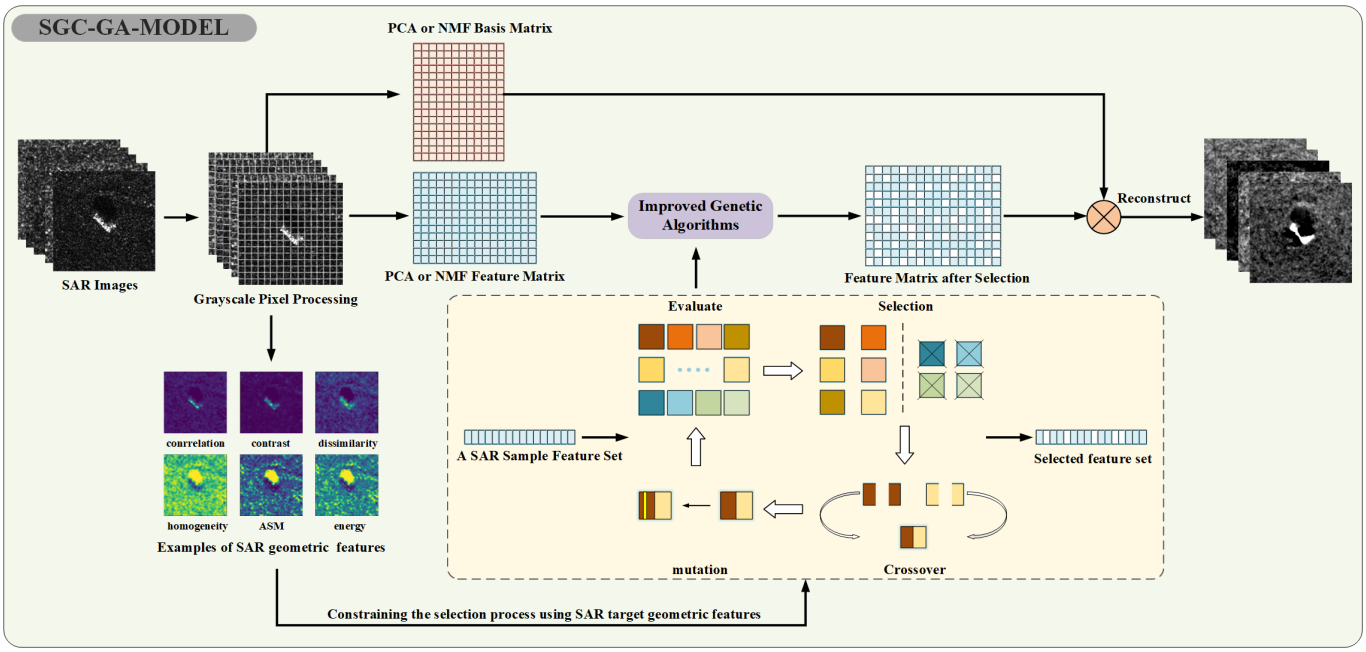


Fig. 2. SGC-GA-Model overall process.

extracted from the grayscale image based on the relationship between the luminance values and the central pixel, and the neighborhood defined by the kernel or window size.

A total of 14 texture features were defined in [48] and in this study, we considered only the following six texture features.

1) *Correlation:* Correlation is a measure of linear correlation between the gray levels of an image, which is used to measure the degree of similarity between the gray levels of an image in the row or column direction.
2) *Contrast:* Contrast measures the spatial frequency of an image, which is able to contrast the central measure of local variations present in an image, and measure the roughness of an image's texture.
3) *Dissimilarity:* Dissimilarity is a linear measure of local variation in an image.
4) *Homogeneity:* Homogeneity can measure the local homogeneity in an image. Meanwhile, it can reflect the homogeneity of the image texture and measure the amount of local variations in the image texture.
5) *ASM:* ASM can measure texture uniformity, i.e., the repetition of pixel pairs.
6) *Energy:* Energy is calculated as the square root of the angular second moment and is a measure of how stable the grayscale variation of the image texture is, reflecting the uniformity of the grayscale distribution and the texture coarseness of the image.

Listed below are the equations illustrating how each feature is calculated:

$$\text{correlation} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}\frac{(i-\mu_i)(j-\mu_j)}{\sqrt{(\sigma_i^2)(\sigma_j^2)}} \tag{1}$$

$$\text{contrast} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}(i-j)^2 \tag{2}$$

$$\text{homogeneity} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}\frac{P(i,j)}{1+(i-j)^2} \tag{3}$$

$$\text{dissimilarity} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}P(i,j)x|i-j| \tag{4}$$

$$\text{ASM} = \sum_{i=0}^{N-1}\sum_{j=0}^{N-1}P(i,j)^2 \tag{5}$$

$$\text{energy} = \sqrt{\sum_{i=0}^{N-1}\sum_{j=0}^{N-1}P(i,j)^2} \tag{6}$$

where N is the number of gray levels, $P(i,j)$ is the normalized value of the gray values at the ith and jth positions of the kernel, summed to 1, μ_i and μ_j are the mean values of the gray values at the different ith and jth positions, respectively, and σ_i^2 and σ_j^2 are the variances of the gray values at the different ith and jth positions, respectively.

The noise of an image is defined as the absolute value of the difference between the gray value of the pixel in the image and the average value of the overall pixel gray value and the SNR is calculated as shown below.

Calculate the noise at each point

$$K(i,j) = P(i,j) - \mu_{i,j} \tag{7}$$

$$\text{SNR} = 10\lg\frac{\sum_{i=0}^{N-1}\sum_{j=0}^{N-1}P(i,j)^2}{\sum_{i=0}^{N-1}\sum_{j=0}^{N-1}K(i,j)^2}. \tag{8}$$

PSNR is defined based on mean square error (MSE) which can be defined as

$$\text{MSE} = \frac{1}{MN}\sum_{i=0}^{M-1}\sum_{j=0}^{N-1}\left[P(i,j)-K(i,j)\right]^2. \tag{9}$$

Then PSNR can be defined as

$$\text{PSNR} = 20\lg\left(\frac{\text{MAX}_{P(i,j)}}{\sqrt{\text{MSE}}}\right). \tag{10}$$

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Save the six features in GLCM with SNR and PSNR features for a total of eight features as constraints.

2) Pattern Feature Selection Based on SAR Target Geometric Features Constraints: This selection algorithm improves on the underlying GA [50], which is a metaheuristic algorithm inspired by the process of biological evolution. The basic elements of the GA are chromosome representation, fitness selection, and heuristic operator setting (selection operator [51], crossover operator [52], and mutation operator [53]). In this study, chromosomes are represented in a binary string format, where each locus (a specific position on the chromosome) has two possible alleles—1 or 0—corresponding to the goal to identify whether one of the traits is adopted or not. And a set of chromosomes represents a set of solutions.

The reasons for choosing the base GA are as follows.

1) The form of chromosome represented by binary characters is suitable for PCA and NMF, which extracts features by matrix transformation, and each point corresponds to a feature dimension, which makes it easy to choose whether or not to use the corresponding feature.
2) The fitness function of the algorithm can be set flexibly according to the problem to be solved.
3) The final output of the GA is the result of the adjustment of the chromosome on the chromosome after adjusting the position of the points, in other words, a set of input into the feature set according to the requirements of the fitness function is selected to produce a new set of better feature subsets. This selection does not destroy the dimensionality of the feature set but rather leaves the unselected features empty, which is equivalent to outputting a superior feature subspace.

Based on the above basic information, our proposed SGC-GA-Model improves the basic GA as follows.

1) Since the problem to be solved is a target recognition problem, our proposed preferred solution introduces the recognition accuracy directly as a fitness function, and guides the selection operator, crossover operator, and mutation operator processes in the direction of improving the accuracy, which is more direct and efficient.
2) Since the basic GA is a stochastic heuristic algorithm, the stochastic selection process in the feature selection process will have problems such as easy to fall into the local optimal solution and long training time. To address the above problems, in the heuristic operator process, SGC-GA-Model introduces a total of eight SAR geometric features of correlation, contrast, similarity, homogeneity, ASM, energy, SNR, and PSNR to constrain the crossover process of the basic GA and enhance the center of gravity.

The crossover process is constrained to enhance the feature expression of the central target object. A threshold τ is set according to the magnitude of SAR geometric features extracted from different datasets, if the difference between the original SAR target image and the reconstructed image after subspace-level preference is less than the set threshold, the chromosomes of the previous generation in the GA can be crossover process, and vice versa, no crossover. This corresponds to the constraint $G = 0$ or 1, respectively, as shown in (11) and (12). The constraints on the feature subspace of the GA are implemented by using the SAR target geometric feature constraints to determine whether or not to crossover, as follows: (13)

$$\begin{cases} \Delta_n(F_o, F_R) < \tau_n \rightarrow \Delta_n(F_o, F_R) = 1 \\ \Delta_n(F_o, F_R) > \tau_n \rightarrow \Delta_n(F_o, F_R) = 0 \end{cases} \tag{11}$$

$$\begin{cases} \Delta_1(F_o, F_R) \text{ and} \cdots \Delta_8(F_o, F_R) = 0 \rightarrow G(F_o, F_R) = 0 \\ \Delta_1(F_o, F_R) \text{ or} \cdots \Delta_8(F_o, F_R) = 1 \rightarrow G(F_o, F_R) = 1 \end{cases} \tag{12}$$

where F_o represents the original SAR target, F_R represents the reconstructed SAR target, τ_n represents the threshold of the nth constraint feature set, Δ_n represents the difference between F_o and F_R for that constraint feature, and G represents the feature constraint sign to decide whether to perform the crossover operator

$$\begin{cases} G(F_o, F_R) = 0 \rightarrow X_{t+1} = X_t \\ G(F_o, F_R) = 1 \rightarrow X_{t+1} = \mathcal{F}_{\text{Crossover}}(X_t) \end{cases} \tag{13}$$

where X represents the current feature layer, t represents the number of iteration rounds, and $\mathcal{F}_{\text{Crossover}}$ represents the crossover operator function.

The improved GA is shown in Fig. 3.

3) Identify Classification and Reconstruction Samples: In the design of this model, an integrated machine learning classifier based on decision tree into the improved GA is introduced, while the output recognition classification accuracy is a function of fitness, and when the algorithm iteration ends, the classification accuracy and the optimal feature subset are output simultaneously.

Using the output optimal feature subset, the unselected feature positions in it are represented by zeros to form a complete feature matrix, i.e., the optimal feature subspace, which is used to achieve image reconstruction in the following steps.

Step 1: Using the feature subset, the PCA and NMF feature matrices are obtained as $Y' \in R^{m\times k}$ and $W' \in R^{m\times k}$, respectively, which are multiplied with the inverse matrices of the base matrices, respectively, to obtain the reconstructed matrices X_{PCA} and X_{NMF}, as shown in the following equations:

$$X_{\text{PCA}} = Y' * V_{\text{PCA}}^{\text{T}} \tag{14}$$

$$X_{\text{NMF}} = W' * H_{\text{NMF}}^{\text{T}}. \tag{15}$$

Step 2: According to the position of the original SAR image, select the features in the specified rows from the reconstructed matrix $X_{\text{PCA}} \in R^{m\times n}$ or $X_{\text{NMF}} \in R^{m\times n}$, two-dimensionalize the original SAR image according to its size, and convert the value of each pixel position into the number of gray levels of the grayscale image for image reconstruction. Fig. 4 shows the comparison between the reconstructed image and the original SAR image after subspace-level preference is performed on the PCA features, and a few samples from the dataset are used as examples here. In the field of SAR target recognition, the target area, shadow area, and background area have different impacts on classification performance [54]. Due to limitations

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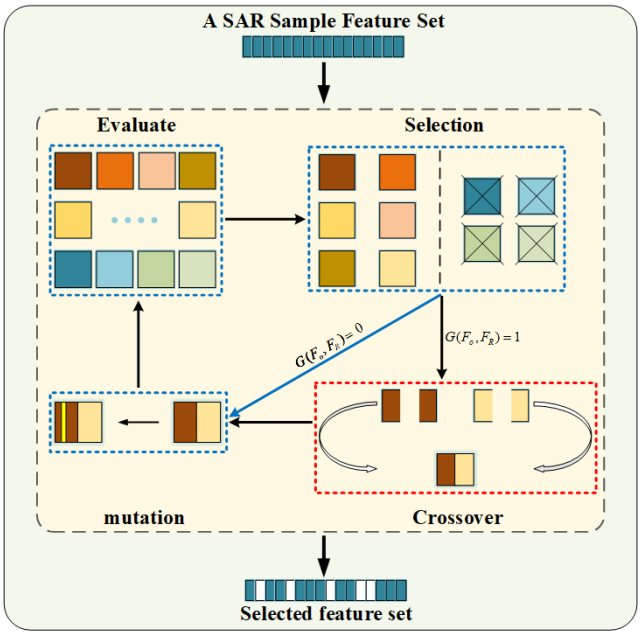


Fig. 3. GA improvement process.

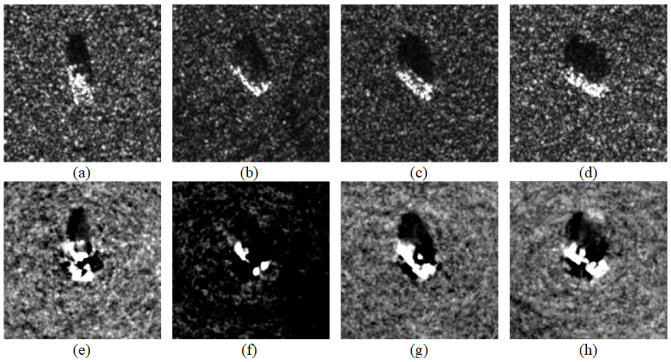


Fig. 4. Original SAR images and the reconstructed images. (a)–(d) Original images of different classes of targets. (e)–(h) Corresponding reconstructed images.

in imaging conditions, SAR target datasets often exhibit data biases, such as background correlation, pseudo-correlation between background clutter properties and target categories, and unstable target features under different imaging conditions [55]. The reconstructed samples show that SGC-GA-Model is able to highlight the feature expression of the central target while suppressing the interference of the background.

C. GFIC-Model

GFIC-Model, a global feature information and local feature information interaction capture model, is a deep feature learning branch with high-level abstraction semantics, which has the ability of multiscale channel feature information interaction and cross-feature mapping layer information interaction, and can realize the integration of global feature information and local feature information.

By extracting different scales of channel feature information from each feature mapping layer, the neighboring preset scales of contextual feature information can be combined more accurately, and cross-channel interaction is constructed by

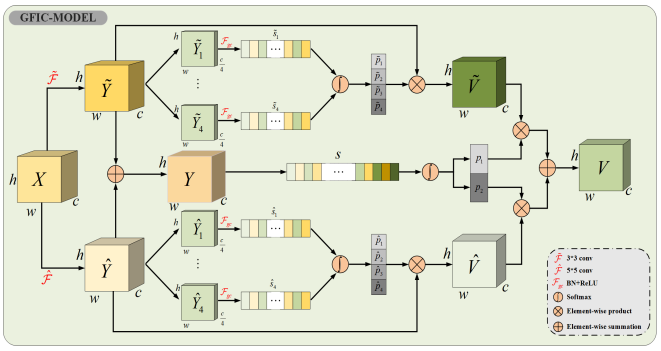


Fig. 5. Structure of GFIC-Model.

extracting the channel attention weights of multiscale feature mapping layers. At the tail end of the cross-feature mapping layer information interaction with the cross-multiscale channel feature information interaction, the feature information weights of different channel outputs are rescaled using the Softmax function to establish remote channel dependencies. The multiscale spatial information and inter-mapping layer information are effectively extracted at a finer granularity level to achieve the integration of global information and capture the interaction between local feature information and global feature information. Meanwhile, the multibranch network structure can further balance the demands of high accuracy and high efficiency, which is an important goal in network development [56]. The structure of GFIC-Model is shown in Fig. 5.

The specific process of GFIC-Model is as follows.

For a given case of feature layer $X \in R^{H\times W\times C}$, two transformations $\tilde{\mathcal{F}} : X \to \tilde{Y} \in R^{H\times W\times C}$ and $\hat{\mathcal{F}} : X \to \hat{Y} \in R^{H\times W\times C}$ are first performed, which consist of a 3×3 convolutional kernel and a 5×5 convolutional kernel, as well as a batch normalization [57] and ReLU function [58], respectively. $\tilde{\mathcal{F}}$ and $\hat{\mathcal{F}}$ were processed to obtain $\tilde{Y}$ and $\hat{Y}$ for the feature layer X, respectively.

For the $\tilde{Y}$ branch, the channel feature information of the input feature mapping layer in a multibranch manner is extracted, and the dimension of the input channel of each branch is $(c/4)$. For each part of it, the channel $\tilde{s}_1 \in R^{(c/4)\times 1}$ is generated by using the global feature information weighting function $\mathcal{F}_{\text{gc}}$ is generated. The $\mathcal{F}_{\text{gc}}$ function is divided into the two processes of generating the information channel and assigning the attention weight to the channel, and first, we use the global averaging pool to embed the global information to generate the information channel $\tilde{s}_{1c}$, as follows:

$$\tilde{s}_{1c} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x_c(i, j). \tag{16}$$

Second, the incentive function is used to assign weights to the information channels to obtain $\tilde{s}_1$, as follows:

$$\tilde{s}_1 = \sigma\left(\mathcal{F}_{\text{FC}}\left(\delta\left(\mathcal{F}_{\text{FC}}\left(\tilde{s}_{1c}\right)\right)\right)\right) \tag{17}$$

where symbol δ denotes the ReLU function, $\mathcal{F}_{\text{FC}}$ denotes the FC layer, and σ denotes the sigmoid activation function. Through the four layers of connectivity in the $\tilde{Y}$-branch, the

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linear feature information between the respective branches can be combined together more effectively, which is conducive to the interaction of the information of different channel dimensions in high, medium, and low. At the same time, the sigmoid activation function can be used to weight the channels after the interaction of the feature information channels, so as to enhance the expression ability of the different dimensional information and extract the feature information more effectively. Due to the symmetry of the branch structure, in the same way, $\tilde{s}_2 \sim \tilde{s}_4$ and $\hat{s}_1 \sim \hat{s}_4$ in $\hat{Y}$ branch can be calculated.

In the $\tilde{Y}$-branch, different spatial scales are adaptively selected using cross-channel soft attention guided by the different feature information channels $\tilde{s}_1 \sim \tilde{s}_4$ generated, with A soft assignment weight is given by the following equation:

$$\tilde{p}_i = \text{Softmax}\left(\tilde{s}_i\right) = \frac{e^{\tilde{s}_i}}{\sum_{i=1}^{4} e^{\tilde{s}_i}} \tag{18}$$

where Softmax is used to obtain the re-calibrated weights of the multiscale channels $\tilde{p}_i$, which contains the position information in the four layers of space in the $\tilde{Y}$-branch and the attention weights in the channel. For each channel, the cross-channel interaction is established in a local way, which achieves the interaction between the local and global channel feature information, and $\tilde{p}_i$ denotes the multiscale channel feature weights after the interaction. In the same way, it can be calculated to get $\hat{p}_i$.

Then, we multiply the multiscale channel feature weights $\tilde{p}_i$ with the corresponding feature mapping layer $\tilde{Y}_i$ to obtain the new feature mapping layer $\tilde{V}$ corresponding to the multiscale channel-by-channel weights, as follows:

$$\tilde{V} = \text{Concat}\left(\left[\tilde{Y}_1 \times \tilde{p}_1, \tilde{Y}_2 \times \tilde{p}_2, \tilde{Y}_3 \times \tilde{p}_3, \tilde{Y}_4 \times \tilde{p}_4\right]\right). \tag{19}$$

Among them

$$\tilde{p}_1 + \tilde{p}_2 + \tilde{p}_3 + \tilde{p}_4 = 1. \tag{20}$$

As shown in the above equation, in the $\tilde{Y}$ branch, the multiscale channel feature information and cross-channel attention are finally integrated into the new feature mapping layer to achieve better feature information interaction between local and global information. In the same way, $\hat{V}$ can be calculated in the $\hat{Y}$ branch.

Based on the completion of information interaction within the different initial feature mapping layers, GFIC-Model then crosses the different initial feature mapping layers and performs information interaction between the different feature mapping layers, and in order to achieve this, the two initial feature mapping layers are first fused as follows:

$$Y = \tilde{Y} + \hat{Y}. \tag{21}$$

The rest of the process is similar to the above analysis, as shown in Fig. 5. First, generate the information channel as follows:

$$s_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} x(i, j). \tag{22}$$

Then, the middle branch and the side branches use the same activation function to create a connected feature channel s, and the Softmax function is utilized to calculate the feature weights p_1 and p_2 between the mapping layers. The final feature mapping layer V is output as follows:

$$V = \tilde{V} \times p_1 + \hat{V} \times p_2. \tag{23}$$

Among them

$$p_1 + p_2 = 1. \tag{24}$$

In such a design, feature information bootstrapping of multiple branches between different feature mapping layers extracted by different convolutional kernel sizes is carried out, and finally feature information aggregation is performed after assigning weights using Softmax function. Different attention to the branches of different feature mapping layers leads to different sizes of the sensory fields of the fusion layer, which enhances the comprehensive expression of global information.

In Fig. 1, the feature dimensions of the shallow, medium, and deep feature layers B_1, B_2, and B_3 extracted by the deep feature extractor after GFIC for feature layers B_1', B_2', and B_3' are 16 384, 8192, and 4096 dimensions, respectively. It can be seen that the feature dimensions of different depths of the deep feature extractor have obvious differences, and this imbalance of feature dimensions may lead to the degradation of recognition accuracy. Meanwhile, deep shallow features with too large dimensions will occupy too large a proportion in the subsequent joint learning process of pattern features and deep features, which may inhibit the full expression of other features, especially for pattern features with relatively small dimensions, and prevent the introduced pattern features from playing the role of positive gain in the joint learning with deep features. In order to solve this problem, we introduce MLFB after GFIC to balance the dimensionality of different feature layers, and downsize the dimensionality of features in each layer to 1024 dimensions by an FC layer.

D. MFJL-Model

In the previous different dimensional feature fusion methods, the add or concat method is often used, as shown in (25) and (26)

$$X_{\text{Classifier}} = X_1 \oplus X_2 \oplus \cdots \oplus X_n \tag{25}$$

where $\oplus$ represents an addition operation

$$X_{\text{Classifier}} = X_1 \copyright X_2 \copyright \cdots \copyright X_n \tag{26}$$

where © denotes the concatenation operation.

However, due to the different correlation between features, different types of features, and large differences in the dimensionality of the feature values, etc., the recognition of feature fusion is not increased, but rather the recognition accuracy will be reduced. Besides, more features are not better, as the dimension of the features increases, the conflict between the features will also increase, and the newly introduced features do not necessarily bring positive gain, but may also be considered as "noise," which will also result in a decrease in recognition accuracy. In order to solve these problems and make the features fully utilized and expressed, MFJL-Model is proposed in this study.

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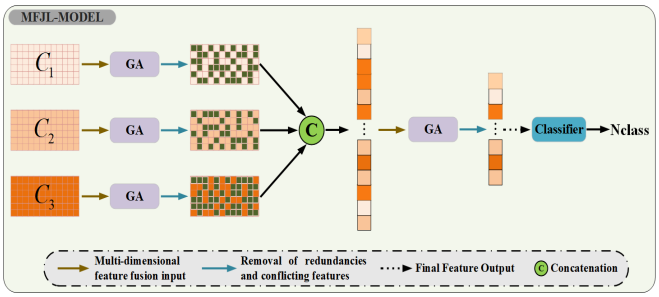


Fig. 6. MFJL-Model overall process.

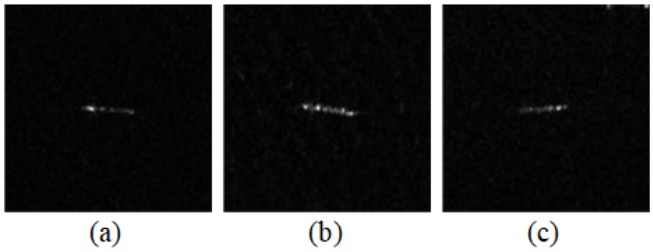


Fig. 7. SAR targets in OpenSARShip-1.0 dataset. (a) Bulk ship. (b) Container ship. (c) Tanker.

The purpose of this model is to fulfill the following functions.

1) Multidimensional Feature Union: In this model, as shown in Fig. 6, C_1, C_2, C_3 are different feature layers. C_1 corresponds to the feature layer B_1'', as follows:

$$C_1 = B_1''. \tag{27}$$

C_2 and C_3 perform the combination of pattern feature layer and depth feature layer; specifically, the C_2 feature layer is a combination of the PCA feature layer A_2' which is learned after branching by SGC-GA-Mechanism and the deep feature layer B_2'' which is extracted by the depth feature extractor of GFIC-Model, as follows:

$$C_2 = A_2' \copyright B_2''. \tag{28}$$

C_3 feature layer is a combination of the NMF feature A_3', which has undergone the learning branch of SGC-GA-Mechanism, and the deep feature layer B_3'', which has been extracted by the deep feature extractor of GFIC-Model as follows:

$$C_3 = A_3' \copyright B_3''. \tag{29}$$

According to the principle of the two joint learning models of SGC-GA-Mechanism and GFIC-Model in the previous section, it can be seen that the SAR image after SGC-GA-Mechanism has enhanced the ability to express the contour of the central target object and the local details of the features, and after GFIC-Model has enhanced the ability to interact between the global feature information and the local feature information of SAR image, and the two complement each other, so as to realize the joint learning of the multidimensional features.

2) Blend of Shallow, Medium, and Deep Multilayer Features: As can be seen from Fig. 6, the features input to the classifier contain some of the features from the C_1, C_2, and C_3 layers of features. Among the pattern features, geometric features, PCA features, and deeper NMF features have been extracted, which correspond to the shallow, medium, and deep features of the deep feature extractor. In SGC-GA-Mechanism, the shallow SAR target geometric features have been used as constraints, so C_1 only retains the shallow deep features, corresponding to the combination of C_2 for the mid-level features and C_3 for the deep features, and all three work together to achieve the final SAR target recognition task.

3) Joint Learning of Multidimensional Features: In previous related work, only simple concating or adding operations have been used in the feature union part, which can lead to a situation where the fused features appear to be dominated by a certain feature. Instead, we put the features into the GA after each feature layer of the joint features C_1, C_2, C_3 after single branch learning to filter the features, also to modify the fitness function of the algorithm to the classification accuracy. Through the iteration of the preference algorithm, the redundant features that cause conflicts in the combined features can be eliminated, and the combination relationship of the features will also be taken into account, which is more targeted than the fusion of simple concating and adding, and it can enhance the expression between different features. After iterating the preference algorithm within each feature layer C_1, C_2, C_3, the three-layer feature union is performed, and the final result of the classifier is obtained after iterating the preference algorithm again, as follows:

$$X_{\text{Classifier}} = \mathcal{F}_{\text{GA}}\left[\mathcal{F}_{\text{GA}}(C_1) \copyright \mathcal{F}_{\text{GA}}(C_2) \copyright \mathcal{F}_{\text{GA}}(C_3)\right] \tag{30}$$

where $\mathcal{F}_{\text{GA}}$ is the basic GA process.

IV. Experiments

All experiments were performed on a computer with Intel[1] Core[2] i5-12400F CPU processor, NVIDIA GeForce RTX 3050 GPU, and 32 g of RAM. All programs were implemented using Python and MATLAB languages, and the deep feature extractor was built on the Pytorch framework.

A. Dataset

1) OpenSARShip Dataset [59]: The OpenSARShip dataset, which aims to develop complex ship detection and classification algorithms in highly disturbed environments, was released by Huang et al. [59] in 2018, and is derived from 41 Sentinel-1 images under different environmental conditions, including VH and VV polarization channels. It has been used by many scholars, e.g., [39], [40]. The distribution of the OpenSARShip-1.0 dataset is shown in Table I. Also, some samples from the OpenSARShip-1.0 dataset are shown in Fig. 7.

2) FUSAR-Ship Dataset [60]: The FUSAR-Ship dataset was acquired by the Gaofen-3 satellite and has a richer variety of ship types compared to the OpenSARShip dataset. As in [40], the ship types used in our experiments are Bulk-Carrier, ContainerShip, Fishing, GeneralCargo, Othercargo, Tanker, and Others, and the same data preprocessing methods

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TABLE I
DISTRIBUTION OF OPENSARSHIP-1.0 DATASET

Category	Training	Test	All
Bulk ship	338	328	666
Container ship	338	808	1146
Tanker	338	146	484

TABLE II
DISTRIBUTION OF FUSAR-SHIP DATASET

Category	Training	Test	All
Container ship	1219	523	1742
General cargo	1205	517	1722
Fishing	1101	473	1574
Tanker	1215	521	1763
Bulk carrier	1150	494	1644
Other cargo	1214	520	1734
Other	1211	521	1732

TABLE III
DISTRIBUTION OF MSTAR-T72VARIANTS DATASET

Category	Training	Test	All
A04	299	274	573
A05	299	274	573
A07	299	274	573
A10	296	271	567
A32	298	274	572
A62	299	274	573
A63	299	274	573
A64	299	274	573

TABLE IV
DISTRIBUTION OF SAR-AIRCRAFT-1.0 DATASET

Category	Training	Test	All
A330	278	31	309
A320/321	1719	52	1771
A220	3270	460	3730
ARJ21	825	362	1187
Boeing737	2007	550	2557
Boeing787	2191	454	2645
Other	3223	1041	4264

and training–testing ratios are applied. The distribution of the FUSAR-Ship dataset is shown in Table II. Also, some samples from the FUSAR-Shipdataset are shown in Fig. 8.

3) MSTAR-T72Variants Dataset [61]: The MSTAR-T72Variants dataset, published in conjunction with the MSTAR dataset, is a fine-grained dataset of the same class of military targets, which contains eight T-72 variants, A04, A05, A07, A10, A32, A62, A63, and A64. These targets are the images of each target in the direction of the radar when it is operated at a variety of different elevation angles. The distribution of the MSTAR-T72Variants dataset is shown in Table III.

4) SAR-AIRcraft-1.0 Dataset [62]: SAR-AIRcraft-1.0 is a high-resolution dataset. The dataset is derived from the Gaofen-3 satellite and contains 4368 images and 16 463 instances of aircraft targets, covering seven fine-grained identification categories, namely A220, A320/321, A330, ARJ21, Boeing 737, Boeing 787, and other. The distribution of the SAR-AIRcraft-1.0 dataset is shown in Table IV.

B. Experiments Settings

In the feature extraction process, the SAR image was resized to 128 × 128 pixels for PCA, NMF, and deep feature extraction processes, respectively. Among them, in the deep feature extraction process, in order to avoid the negative effects of the small number of samples and the imbalance in the number between the species, simple data enhancement means such as random cropping and random flipping were adopted for the OpenSARShip-1.0 and FUSAR-Ship datasets.

For deep feature extraction on the four datasets, the Adaptive Moment Estimation (Adam) optimizer [73] was used, and ResNet-18 was used for the base network for 100 epochs of training, with the training batch size set to 64 and the learning rate set to 0.0005.

In the proposed overall framework, SGC-GA-Model and MFJL-Model for the GA model, the base parameters are set as follows: crossover probability is 0.7, variance probability is 0.1, and the number of individuals in the population is 25, where SGC-GA-Model for the constraint thresholds is set by the magnitude of the constraints, and the constraint thresholds for different features in this article are set to 5% of the feature value of the original SAR image.

For the feature input to classifier session at the end of the framework, we used lightGBM [74], a decision tree-based integrated classifier first released by Microsoft in 2017, where the maximum depth of each tree is set to 5 and the number of leaf nodes per tree is set to 96.

C. Evaluation Metric

In order to demonstrate the effectiveness of the method proposed in this article, we use some evaluation metrics to define the experimental results. There are four main methods: true positives (TP), true negatives (TN) and false positives (FP), false negatives (FN). Accuracy (%) is defined as

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}. \tag{31}$$

Other metrics were also used here, including recall (%) and precision (%), which are defined as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{32}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{33}$$

Also, confusion matrix and T-SNE data downscaling visualization were also adopted to demonstrate the accuracy of target recognition.

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TABLE V
ACCURACY OF FJL-FRAMEWORK ON DIFFERENT TYPES OF SAR TARGET RECOGNITION DATASETS

Dataset	Accuracy(%)	Precision(%)	Recall(%)
OpenSARShip-1.0	79.02±0.15	73.97±0.54	78.38±0.49
FUSAR-Ship	87.59±0.17	87.86±0.39	88.54±0.37
MSTAR-T72Variants	98.03±0.11	97.99±0.32	97.99±0.41
SAR-AIRcraft-1.0	99.51±0.13	99.57±0.39	99.61±0.47

TABLE VI
COMPARISON OF OPENSARSHIP-1.0 AND FUSAR-SHIP WITH OTHER TARGET RECOGNITION METHODS

Feature	Method	OpenSARShip-1.0			FUSAR-Ship		
		Accuracy(%)	Precision(%)	Recall(%)	Accuracy(%)	Precision(%)	Recall(%)
Handcrafted Feature	SVM [63]	59.34	53.21	60.78	74.19	73.25	74.19
	KNN [64]	63.89	61.14	64.97	68.87	69.1	68.87
	Decision Tree [65]	45.48	43.26	48.11	56.01	55.76	56.01
	Random Forest [66]	52.86	47.51	55.34	72.46	72.08	72.46
Deep Feature	ResNet-18 [47]	72.69±0.65	66.75±1.84	67.95±1.78	78.78±0.31	78.71±0.26	78.63±0.23
	ResNet-34 [47]	71.22±1.07	67.75±1.71	67.97±2.23	78.34±0.24	78.37±0.19	78.18±0.21
	ResNet-50 [47]	69.73±0.78	65.74±1.95	67.55±1.89	80.28±0.35	80.21±0.31	80.15±0.26
	ResNet-101 [47]	69.03±0.61	63.56±2.14	66.16±2.45	79.76±0.27	80.07±0.35	79.56±0.31
	ResNet-152 [47]	68.76±1.03	63.54±1.83	65.87±2.06	79.41±0.29	79.85±0.24	79.18±0.27
	ResNext-50-32x4d [67]	68.17±0.83	63.75±2.06	69.34±2.31	79.60±0.35	79.38±0.37	79.40±0.37
	ResNext-101-32x8d [67]	67.54±0.67	65.22±1.95	68.67±2.56	79.72±0.28	79.93±0.32	79.21±0.34
	Wide-ResNet-50_2 [67]	70.45±1.09	67.91±1.74	69.78±2.21	80.53±0.37	80.83±0.37	80.59±0.32
	Wide-ResNet-101_2 [67]	70.28±0.81	66.91±1.73	68.34±2.57	79.86±0.57	80.18±0.48	80.07±0.54
	VGG-11 [68]	72.23±0.77	68.57±1.41	70.35±0.87	80.41±0.16	80.46±0.23	80.29±0.22
	VGG-13 [68]	72.85±0.85	70.21±1.32	69.64±1.22	81.03±0.21	80.92±0.27	80.88±0.27
	VGG-16 [68]	72.23±1.27	67.75±2.43	68.32±3.76	80.25±0.24	80.12±0.25	80.10±0.34
	VGG-19 [68]	71.14±1.39	67.17±4.31	69.67.±4.97	80.36±0.31	80.23±0.61	80.22±0.49
	DenseNet-121 [69]	73.71±0.74	71.24±2.17	65.93±2.96	82.34±0.39	82.48±0.41	82.12±0.39
	DenseNet-161 [69]	74.26±0.67	69.56±2.01	67.86±3.21	82.57±0.26	82.42±0.33	82.43±0.43
	DenseNet-169 [69]	72.54±0.72	66.64±1.79	68.61±2.65	81.87±0.33	82.18±0.38	82.67±0.46
	DenseNet-201 [69]	73.24±0.81	70.86±1.84	67.91±3.34	82.29±0.28	82.22±0.41	82.16±0.37
	SqueezeNet-v1.0 [70]	70.83±1.15	65.09±1.71	66.15±1.67	78.58±0.37	78.56±0.45	78.66±0.44
	MobileNet-v2 [71]	69.11±0.93	62.21±2.35	66.2±3.57	73.84±0.39	74.27±1.34	73.71±2.45
Feature Fusion	Wang et al. [72]	69.27±0.27	58.72±4.76	57.72±1.37	80.02±0.46	80.32±0.67	80.09±0.47
	Hou et al. [60]	67.41±1.13	69.44±2.42	69.33±2.00	71.89±0.54	72.19±0.59	71.96±0.56
	Huang et al. [27]	74.98±1.46	69.56±2.38	74.74±1.60	83.19±0.31	83.34±0.31	83.19±0.41
	Zhang et al. [39]	78.15±0.57	72.42±1.06	77.87±1.14	86.69±0.47	86.54±0.50	86.62±0.51
	Zheng et al. [40]	78.60±0.21	73.78±0.62	**79.37±0.58**	87.23±0.26	86.89±0.45	86.67±0.43
Feature Joint Learning	FJL-Framework	**79.02±0.15**	**73.97±0.54**	78.38±0.49	**87.59±0.17**	**87.86±0.39**	**88.54±0.37**

D. Recognition Experiments Results and Analysis

1) Target Recognition Performance on Different Datasets: Table V shows the recognition accuracies of the model proposed in this article on OpenSARShip-1.0, FUSAR-Ship datasets, and two fine-grained datasets, MSTAR-T72Variants and SAR-AIRcraft-1.0, where ten experiments are performed on each dataset to obtain the mean and standard deviation. As shown in Table V, OpenSARShip-1.0 recognition accuracy is 79.02%, FUSAR-Ship recognition accuracy is 87.59%, MSTAR-T72Variants recognition accuracy is 98.03%, and SAR-AIRcraft-1.0 recognition accuracy is 99.51%. The model proposed in this article shows excellent performance on different types of SAR target recognition tasks, such as ships, tanks, and aircraft, and also achieves good results in the performance of fine-grained tasks.

2) Performance and Analysis of Target Recognition With Different Methods: Table VI shows the comparison of quantitative evaluation metrics between the method proposed in this article and common traditional features such as SVM [63], KNN [64], decision tree [65], random forest [66], and common

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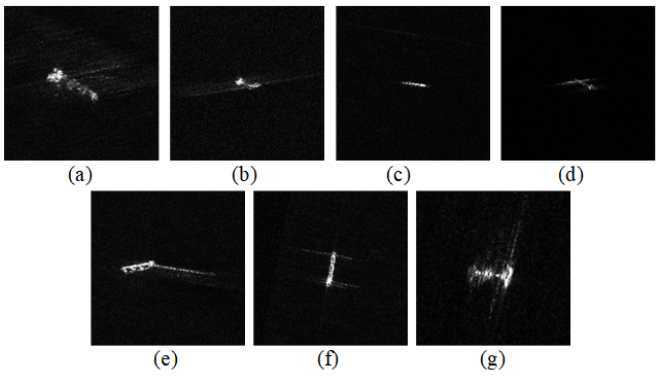


Fig. 8. SAR targets in FUSAR-Ship dataset. (a) Container ship. (b) General cargo. (c) Fishing. (d) Tanker. (e) Bulk carrier. (f) Other cargo. (g) Others.

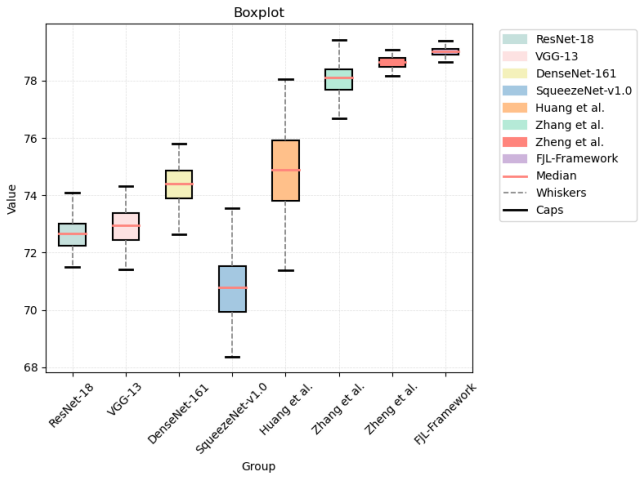


Fig. 9. OpenSARShip-1.0: Comparison of the mean accuracy and standard deviation of the FJL-Framework-based feature joint learning method with other typical methods using box plots.

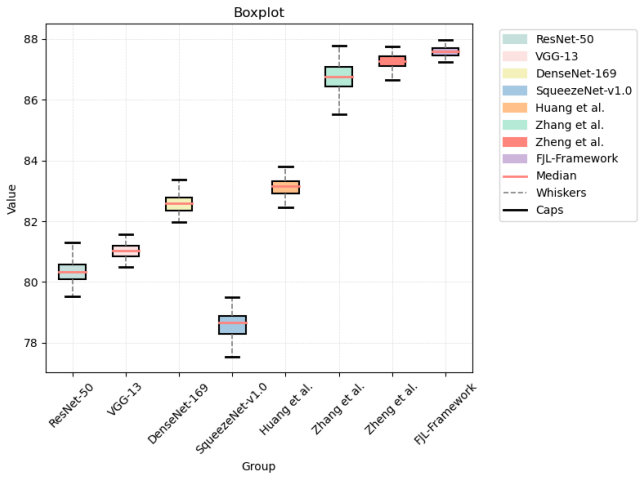


Fig. 10. FUSAR-Ship: Comparison of the mean accuracy and standard deviation of the FJL-Framework-based feature joint learning method with other typical methods using box plots.

deep network models such as ResNet series [47], [67], Vgg series [68], densenet series [69], and Lightweight Networks [70], [71], where ten experiments are conducted to obtain the mean and standard deviation.

As can be seen in Table VI, among the manual feature methods, KNN method achieved the highest accuracy of 63.89% in OpenSARShip-1.0 and SVM method achieved the highest accuracy of 74.19% in FUSAR-Ship. Compared to the manual feature methods, the deep feature methods clearly have superior recognition accuracy, with the DenseNet-161 model achieving the highest accuracy of 74.26% and 82.57% on both OpenSARShip-1.0 and FUSAR-Ship datasets, respectively. As can be seen from Table VI, the performance of the method being proposed in this article has a significant improvement over the models with both manual and deep features.

Meanwhile, two major datasets OpenSARShip-1.0 and FUSAR-Ship are used as examples to conduct comparative experiments on recent feature fusion-related work, in which the MFCFNet proposed by Zheng et al. [40] achieves excellent recognition accuracies of 78.60% and 87.23% in these two datasets, respectively. Our proposed method has more accurate recognition ability and achieves the highest recognition accuracy of 79.02% and 87.59% on both datasets, and the proposed method does not introduce the traditional pattern features into the network training and destroy its original features like other related works, and proposed method also eliminates the redundant irrelevant and conflicting features. which will be shown in detail in the experimental results later.

Further, taking the two major datasets, OpenSARShip-1.0 and FUSAR-Ship, as examples, the better models were selected from the different series of depth models, the better models were selected from the feature fusion work, and our method to plot the box plots, and further visualize the "Accuracy $\pm$ Standard Deviation" of the experimental results. The boxes of the box-and-line plot represent the middle 50% range of the data, and we use the horizontal line in the box to represent the mean of the accuracy, and the two lines on the outside of the box (also known as Whiskers) to represent the overall range of fluctuations in the data, i.e., the standard deviation. As shown in Figs. 9 and 10, the proposed method in this article has the highest recognition accuracy and smaller range of fluctuation in the recognition task for both OpenSARShip-1.0 and FUSAR-Ship datasets, and the stability of the method is more superior due to the fact that the joint learning method retains the optimal features, and the stability of the features is higher.

Then, taking the OpenSARShip-1.0 dataset as an example, confusion matrices were generated to display the better models from different series of deep models, feature fusion works, and the proposed method, as shown in Fig. 11. It is obvious that the proposed method has more balanced recognition ability for each class in the recognition task compared to other methods, which is because the joint learning will allow the traditional pattern features and deep features to perform their respective superiority, which can better balance the feature requirements between the classes, and avoid the great deviation of recognition ability between different classes.

In order to further illustrate the effectiveness of the feature joint learning method proposed by ours as well as the performance level of the proposed improvement model in fine-grained tasks, we conducted extended experiments on two fine-grained datasets, MSTAR-T72Variants and SAR-AIRcraft-1.0, and the results are shown in Table VII.

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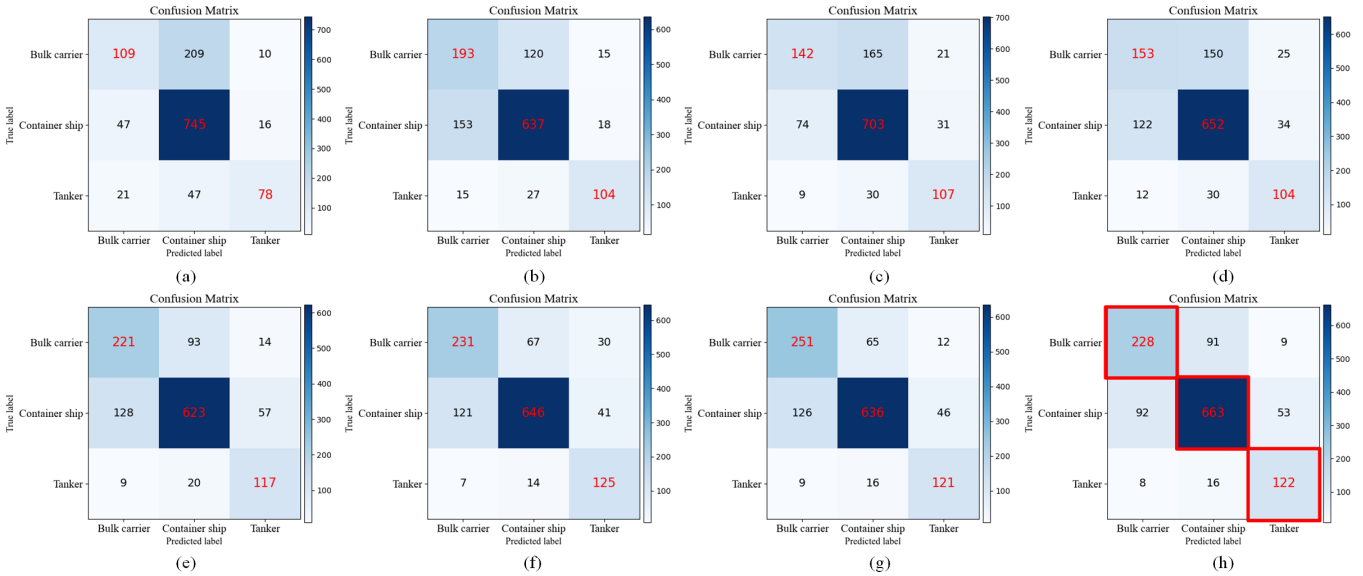


Fig. 11. Comparison of the confusion matrix of the FJL-Framework-based feature joint learning method with other typical methods. (a) ResNet-18. (b) VGG-13. (c) DenseNet-161. (d) SqueezeNet-v1.0. (e) Huang et al. (f) Zhang et al. (g) Zheng et al. (h) FJL-Framework.

TABLE VII
COMPARISON OF MSTAR-T72VARIANTS AND SAR-AIRCRAFT-1.0 WITH OTHER TARGET RECOGNITION METHODS

Feature	Method	MSTAR-T72Variants			SAR-AIRcraft-1.0		
		Accuracy(%)	Precision(%)	Recall(%)	Accuracy(%)	Precision(%)	Recall(%)
Handcrafted Feature	SVM [63]	59.02	65.68	66.77	95.21	96.44	96.52
	KNN [64]	58.57	60.52	61.59	94.77	95.21	95.33
	Decision Tree [65]	56.51	57.41	58.33	92.14	92.91	93.04
	Random Forest [66]	54.52	63.37	64.45	88.85	89.67	89.79
Deep Feature	ResNet-18 [47]	95.57±0.71	95.76±1.73	95.57±1.77	95.42±0.82	96.60±1.67	94.89±1.87
	ResNet-34 [47]	94.61±1.32	94.73±1.93	94.61±1.92	95.84±0.97	96.58±1.86	95.33±2.04
	ResNet-50 [47]	91.50±1.01	91.56±1.68	91.51±1.74	94.66±1.24	95.63±1.89	94.13±1.75
	ResNet-101 [47]	83.87±1.31	84.44±1.95	83.88±1.91	91.32±1.45	91.66.±1.73	89.17±1.98
	ResNet-152 [47]	76.61±1.22	78.01±2.13	76.63±1.67	92.35±1.12	89.34±2.31	92.99±2.04
	ResNext-50-32x4d [67]	85.88±0.93	86.13±1.64	85.90±2.31	94.08±1.37	94.45±1.74	93.22±2.31
	ResNext-101-32x8d [67]	88.12±0.86	88.57±1.77	88.14±2.09	93.60±0.97	94.51±1.67	93.91±2.11
	Wide-ResNet-50_2 [67]	94.34±0.91	94.59±1.91	94.34±2.44	95.08±0.89	95.72±1.86	94.86±2.56
	Wide-ResNet-101_2 [67]	89.54±1.23	90.29±1.86	89.55±2.21	92.63±1.15	94.45±1.99	89.61±2.13
	VGG-11 [68]	93.24±0.67	93.39±1.52	93.25±1.12	97.51±0.75	97.79±1.47	97.42±1.52
	VGG-13 [68]	94.29±0.94	94.59±1.61	94.30±1.36	97.87±0.81	98.33±1.67	97.20±1.63
	VGG-16 [68]	93.38±1.26	93.50±2.57	93.38±1.95	97.17±1.41	97.96±2.77	95.95±2.06
	VGG-19 [68]	94.24±1.07	94.38±3.64	94.25±3.89	95.96±1.22	95.91±3.25	95.35±3.91
	DenseNet-121 [69]	92.10±0.79	92.25±2.35	92.11±2.63	91.92±0.87	92.41±2.41	90.23±2.69
	DenseNet-161 [69]	94.52±0.96	94.71±1.77	94.52±3.11	96.27±0.91	95.43±1.78	96.16±2.97
	DenseNet-169 [69]	93.74±1.13	93.76±1.96	93.75±2.56	91.26±1.04	92.77±1.64	88.85±2.31
	DenseNet-201 [69]	90.68±1.33	90.81±2.14	90.69±3.34	93.14±1.21	93.49±1.97	92.27±2.54
	SqueezeNet-v1.0 [70]	90.32±1.24	90.53±1.89	90.33±1.71	87.68±1.61	63.89±1.75	68.53±1.89
	MobileNet-v2 [71]	92.74±1.15	93.27±1.99	92.75±3.62	92.59±1.37	93.37±2.06	91.16±3.61
Feature Fusion	Wang et al. [72]	92.78±0.97	89.71±3.34	88.21±2.65	93.41±0.94	92.24±2.61	91.97±1.73
	Hou et al. [60]	90.12±1.32	91.32±2.55	90.78±2.11	91.89±1.04	92.03±2.23	91.92±1.44
	Huang et al. [27]	94.93±1.12	93.56±1.61	93.61±2.34	96.47±0.97	96.23±1.86	96.35±1.65
	Zhang et al. [39]	97.21±0.87	97.32±1.22	97.24±1.79	99.13±0.78	98.96±1.46	99.07±1.37
	Zheng et al. [40]	97.62±0.94	97.64±0.97	97.63±1.21	99.21±0.67	99.28±1.06	99.34±1.17
Feature Joint Learning	FJL-Framework	**98.03±0.11**	**97.99±0.32**	**97.99±0.41**	**99.51±0.13**	**99.57±0.39**	**99.61±0.47**

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TABLE VIII
VARIATION OF FEATURE DIMENSIONS FOR DIFFERENT FEATURE LAYERS IN FJL-FRAMEWORK

	OpenSAR Ship-1.0	FUSAR-Ship	MSTAR-T72Variants	SAR-AIR craft-1.0
$A_2 \rightarrow A_2^{'}$	100 51	100 54	100 68	100 71
$A_3 \rightarrow A_3^{'}$	100 64	100 61	100 67	100 59
$B_1^{''} \rightarrow C_1$	1024 525	1024 529	1024 513	1024 536
$A_2^{'} © B_2 \rightarrow C_2$	1075 531	1078 535	1092 530	1095 554
$A_3^{'} © B_3 \rightarrow C_3$	1088 534	1085 527	1091 519	1083 539
$C_1 © C_2 © C_3 \rightarrow end$	1590 813	1591 763	1562 761	1629 765

TABLE IX
QUANTITATIVE COMPARISON OF PATTERN FEATURE PROCESSING USING DIFFERENT METHODS

Dataset	Method	$A_2^{'}$dimension	$A_3^{'}$dimension	Accuracy(%)
OpenSAR Ship-1.0	NONE	100	100	73.77±0.22
	GA	67	77	76.82±0.41
	SGC-GA	**51**	**64**	**79.02±0.15**
FUSAR-Ship	NONE	100	100	81.54±0.26
	GA	73	75	84.61±0.52
	SGC-GA	**54**	**61**	**87.59±0.17**
MSTAR-T72Variants	NONE	100	100	94.32±0.51
	GA	81	82	96.94±1.31
	SGC-GA	**68**	**67**	**98.03±0.11**
SAR-AIR craft-1.0	NONE	100	100	96.24±0.15
	GA	85	79	98.89±1.12
	SGC-GA	**71**	**59**	**99.51±0.13**

TABLE X
ABLATION EXPERIMENTS WITH THE GFIC-MODEL

Dataset	Method	Accuracy(%)
OpenSARShip-1.0	NONE	77.13±0.44
	GFIC	**79.02±0.15**
FUSAR-Ship	NONE	85.31±0.37
	GFIC	**87.59±0.17**
MSTAR-T72Variants	NONE	96.28±0.32
	GFIC	**98.03±0.11**
SAR-AIRcraft-1.0	NONE	98.19±0.22
	GFIC	**99.51±0.13**

TABLE XI
QUANTITATIVE COMPARISON OF DEEP FEATURE PROCESSING USING DIFFERENT METHODS

Dataset	Method	Accuracy(%)
OpenSARShip-1.0	Non-local	78.23±0.52
	SE	77.37±0.33
	CBAM	78.74±0.61
	GFIC(Ours)	**79.02±0.15**
FUSAR-Ship	Non-local	86.79±0.45
	SE	86.12±0.29
	CBAM	87.22±0.47
	GFIC(Ours)	**87.59±0.17**
MSTAR-T72Variants	Non-local	97.54±0.37
	SE	97.32±0.24
	CBAM	97.68±0.32
	GFIC(Ours)	**98.03±0.11**
SAR-AIRcraft-1.0	Non-local	98.54±0.49
	SE	98.27±0.26
	CBAM	98.89±0.43
	GFIC(Ours)	**99.51±0.13**

TABLE XII
QUANTITATIVE COMPARISON OF DIFFERENT FEATURE FUSION METHODS

Dataset	Method	Dimension	Accuracy(%)
OpenSARShip-1.0	Concat(None)	3187	72.83±1.69
	Add	1000	75.13±1.42
	MFJL(Ours)	**813**	**79.02±0.15**
FUSAR-Ship	Concat(None)	3183	79.53±1.43
	Add	1000	83.36±1.35
	MFJL(Ours)	**763**	**87.59±0.17**
MSTAR-T72Variants	Concat(None)	3207	93.59±1.57
	Add	1000	96.44±1.45
	MFJL(Ours)	**761**	**98.03±0.11**
SAR-AIRcraft-1.0	Concat(None)	3202	97.52±1.06
	Add	1000	98.69±0.77
	MFJL(Ours)	**765**	**99.51±0.13**

It can be clearly seen that the proposed method obtains near-perfect recognition accuracy on these two datasets, which is far superior to the comparison models. The fine-grained recognition task is more challenging compared to the ordinary recognition task. Due to the more similar contours, details, physical attributes, etc., of the classes to be recognized, there is a greater need for innovative design of the feature extraction as well as the utilization process, whereas the proposed method of joint learning of features can continuously learn different types of features with the aim of enhancing the recognition capability by retaining the more discriminative combinations of features and eliminating the redundant irrelevant or conflicting features, and show excellent performance in fine-grained recognition tasks.

3) Rejection of Redundant Features: Table VIII demonstrates the ability of the model proposed by ours in terms of redundant or contradictory feature rejection, where the rows of the table represent the before and after feature dimensionality change of a particular feature layer for different datasets. Rows 2–3 of the table show the before and after dimensionality changes of the two feature layers PCA and NMF after pattern feature extraction, rows 3–5 show the before and after feature dimensionality changes of each feature layer C_1, C_2,

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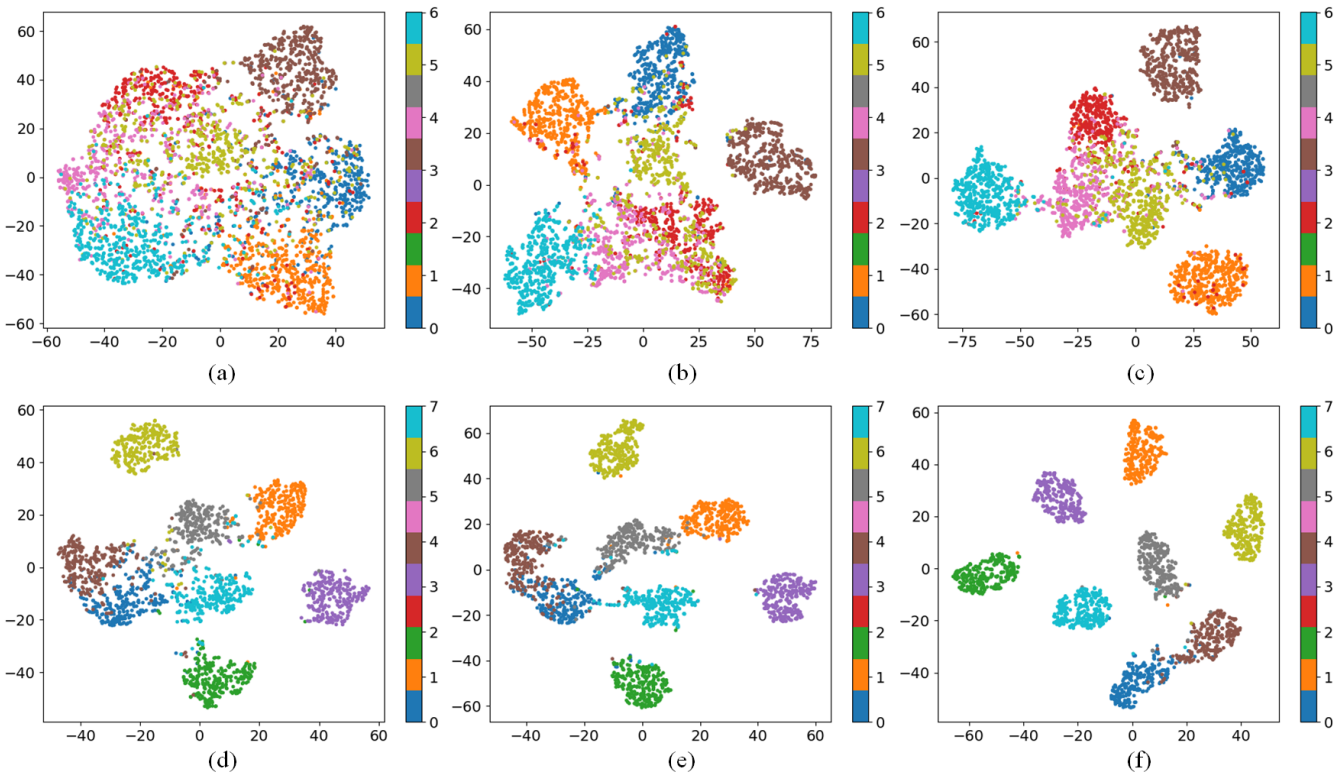


Fig. 12. Final T-SNE visualization results for different types of feature fusion approaches. (a) FUSAR-Ship (Concat). (b) FUSAR-Ship (Add). (c) FUSAR-Ship (MFJL-Model). (d) MSTAR-T72Variants (Concat). (e) MSTAR-T72Variants (Add). (f) MSTAR-T72Variants (MFJL-Model).

C_3 from the shallow to the deeper feature layers, and row 6 shows the before and after feature dimensionality changes of the final C_1, C_2, C_3 fusion.

As can be seen from Table VIII, almost before and after different feature layers in each dataset, the feature dimension of the feature layer decreases by half after the method proposed by ours, and about half of the redundant irrelevant, or conflicting features are eliminated in the various branches of the FJL-Framework, for example, in the last feature layer in the four datasets before inputting into the classifier decreased by 48.87%, 52.04%, 51.28%, and 51.61%, respectively. The preferred combination of features with the best recognition ability retained based on the features of the recognized target is aggregated into the final feature layer, which finally completes the SAR target recognition task. According to the needs of the recognition task, the adaptive feature culling for the purpose of improving the recognition ability cannot be done by other methods, which also reflects the importance of joint learning.

E. Ablation Experiments and Analysis

In order to validate the effectiveness of the different models in the joint learning framework proposed in this article, we conducted further studies of ablation experiments on the OpenSARShip-1.0 and FUSAR-Ship datasets, as well as on two fine-grained datasets, MSTAR-T72Variants and SAR-AIRcraft-1.0.

1) Ablation Experiment of SGC-GA-Model: SGC-GA-Model is a separate learning branch of traditional pattern features, whose effectiveness will illustrate in terms of the ability to eliminate redundant irrelevant and conflicting features as well as the recognition accuracy. Simultaneously, the experimental results also validated the constraint effect

TABLE XIII
COMPARISON OF ABLATION EXPERIMENTS WITH DIFFERENT MODELS

Dataset	SGC-GA	GFIC	MFJL	Accuracy(%)
OpenSAR Ship-1.0	✘	✘	✘	68.73±0.97
	✘	✔	✔	73.77±0.22
	✔	✘	✔	77.13±0.44
	✔	✔	✘	72.83±1.69
	✔	✔	✔	**79.02±0.15**
FUSAR-Ship	✘	✘	✘	77.25±0.82
	✘	✔	✔	81.54±0.26
	✔	✘	✔	85.31±0.37
	✔	✔	✘	79.53±1.43
	✔	✔	✔	**87.59±0.17**
MSTAR-T72Variants	✘	✘	✘	89.75±0.71
	✘	✔	✔	94.32±0.51
	✔	✘	✔	96.28±0.32
	✔	✔	✘	93.59±1.57
	✔	✔	✔	**98.03±0.11**
SAR-AIR craft-1.0	✘	✘	✘	92.34±0.91
	✘	✔	✔	96.24±0.15
	✔	✘	✔	98.19±0.22
	✔	✔	✘	97.52±1.06
	✔	✔	✔	**99.51±0.13**

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of SAR target-related features that were introduced separately. As shown in Table IX, columns 3–4 show the feature dimensions of the extracted PCA pattern features and NMF pattern features after processing by different methods, respectively, and column 5 shows the recognition accuracies of the final joint learning corresponding to different methods. The SGC-GA-Model proposed by ours compared with the method without any processing (NONE) and the basic GA that does not incorporate any constraints related to SAR targets' geometric features.

By comparing the results of the recognition tasks on four datasets, it can be observed that incorporating optimization algorithms to traditional pattern features can improve recognition performance to some extent and eliminate redundant features. However, after carefully designing and incorporating SAR target geometric features constraints, the SGC-GA-Model can further select feature combinations with stronger recognition ability for the specific task of SAR target recognition. This not only improves recognition accuracy but also reduces the feature dimension. Taking the OpenSARShip-1.0 dataset as an example, after introducing the SGC-GA-Model, the feature dimensions of A_2' and A_3' decreased from 100 to 49 and 36, respectively, resulting in a 5.25% improvement in overall recognition accuracy.

2) Ablation Experiment of GFIC-Model: The GFIC-Model is a separate learning branch for deep features, and its effectiveness will be demonstrated from the perspective of target recognition ability, as shown in Table X. After introducing the GFIC-Model, there was an improvement of around 2% in the experiments on all four datasets. The GFIC-Model strengthens the connection between the center target object and the global background in SAR images, allowing for the comprehensive utilization of local information of target objects and global background information, which can enhance SAR target recognition tasks.

Furthermore, a comparison was made between GFIC-Model and the attention models Non-local and SE used in related works. As shown in Table XI, it can be seen that the proposed method exhibits better recognition accuracy compared to the Non-local [75], SE [76], and CBAM [77] models. This is attributed to the ability of the proposed method to more effectively extract multiscale spatial information and interlayer information, allowing for the capture of the interaction between local feature information and global feature information. As a result, GFIC-Model achieves superior recognition performance.

3) Ablation Experiment of MFJL-Model: MFJL-Model is the last step of the whole FJL-Framework, which fuses the shallow to deep feature layers of C_1, C_2, and C_3 and finally inputs them into the classifier. Unlike other feature fusion-related work where different features are directly concatenated or added together, MFJL-Model uses an optimization algorithm with recognition accuracy as the objective function as the "fusion agent" for the different feature layers in the FJL-Framework. This model can reduce a large number of redundant irrelevant and conflicting features in the feature combinations obtained from different learning branches, and improve the recognition accuracy while reducing the dimensionality of the features.

As shown in Table XII, a comparison was made between the direct concatenation and addition methods. Due to the varying feature dimensions across different layers, direct addition was not possible. Therefore, PCA was used to reduce the dimensionality of features from each layer to 1000 dimensions before adding them up. Obviously, compared to the other two methods, MFJL-Model achieves the highest accuracy with lower feature dimensions.

Further, taking FUSAR-Ship and MSTAR-T72Variants datasets as examples, the recognition results of three different methods were visualized using T-SNE clustering, as shown in Fig. 12. The visualization results can more intuitively reflect the strength of the recognition ability, with the MFJL-Model method resulting in stronger separation between the individual classes and more obvious clustering effects.

Finally, a comparison of the different model ablation experiments is shown in Table XIII. The highest SAR target recognition accuracy is achieved by the joint learning of features by the three models.

V. Conclusion

In this article, a new joint learning FJL-Framework for multidimensional features of SAR targets is proposed, involving three new models, such as the subspace-level feature preference model based on the constraints of SAR targets' geometric features (SGC-GA-Model), the model of capturing the interaction between global and local feature information, and the joint multidimensional feature learning model (MFJL-Model). These three models make the traditional pattern features and deep features with advanced abstract semantics first complementarily select single-branch features, and then jointly learn to be applied to the SAR target recognition task, which strengthens the efficient joint use of the features, and the ablation experiments of all the three models are investigated in the experimental chapters.

Finally, the main experiments were conducted on two datasets, OpenSARShip-1.0 and FUSAR-Ship, and supplementary experiments on two fine-grained datasets, MSTAR-T72Variants and SAR-AIRcraft-1.0. In the above four datasets containing SAR target types such as ships, tanks, and aircraft, the FJL-Framework proposed in this article is significantly better than other methods, and can achieve the SAR target identification task more reliably, efficiently, and purposefully.

Our future work is as follows.

1) Study the application of more diverse traditional pattern features, and select more targeted traditional pattern features for different recognition tasks and recognition scenarios.
2) To study more diverse forms of joint feature learning, to find more efficient joint models, to apply joint learning of multiple forms of features to recognition tasks, and to improve the model based on this.
3) Explore the commonalities between traditional pattern features and deep features with high-level abstract semantics, and explore a set of exceptional public features as base features for recognition tasks.

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4) Further compression of the model size is achieved on the basis of reduced feature redundancy, which will save storage space.

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